UC3BPR201

Introduction to Data Analysis

from raw data to insight

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Objectives

- To understand data types
- To understand data distribution
- To understand basic descriptive statistics
- Charts and Graphs

What is data?

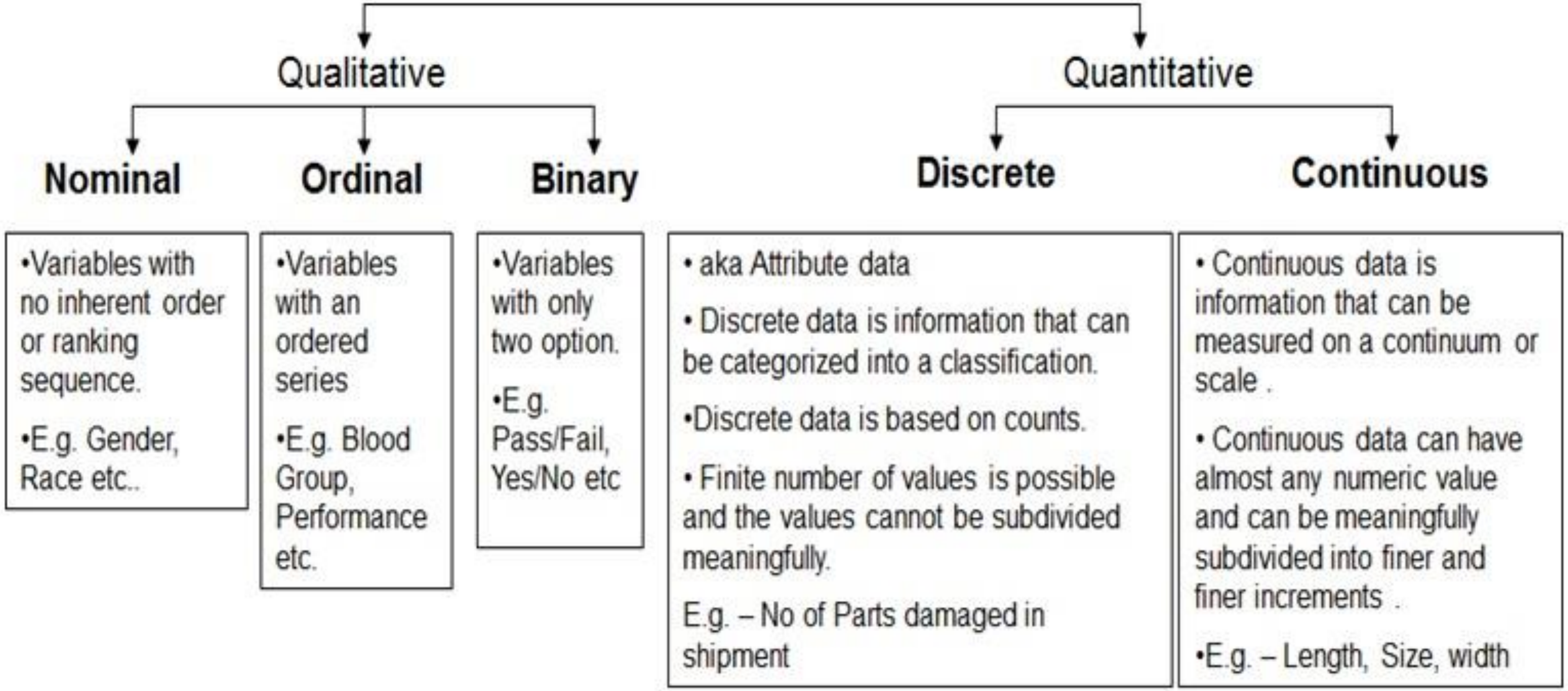
- Data
  - Primary
    - Quantitative
      - Numeric quantity
      - Mostly structured
      - Easy to analyze statistically
    - Qualitative
      - Mostly attributes/properties
      - Describe quality
      - Used to develop initial understanding
  - Secondary

- In a cybersecurity or digital forensic context:

- Quantitative Data (Numerical) is vital for anomaly detection. For example, if the "Number of failed logins" spikes from 10 to 1,000, an automated alert is triggered. It also aids in statistical modeling and evaluating breach probabilities.

- Qualitative Data (Categorical) is vital for Threat Intelligence and Classification. Recognizing the Attack Vector (e.g., Phishing vs. SQL Injection) helps security teams decide how to respond effectively.

What is data?


Qualitative
Quantitative
Nominal
Ordinal
Binary
Discrete
Continuous
•Variables with no inherent order or ranking sequence.
•E.g. Gender, Race etc..
•Variables with an ordered series
•E.g. Blood Group, Performance etc.
•Variables with only two option.
•E.g. Pass/Fail, Yes/No etc
• aka Attribute data
• Discrete data is information that can be categorized into a classification.
•Discrete data is based on counts.
• Finite number of values is possible and the values cannot be subdivided meaningfully.
E.g. – No of Parts damaged in shipment
• Continuous data is information that can be measured on a continuum or scale .
• Continuous data can have almost any numeric value and can be meaningfully subdivided into finer and finer increments .
•E.g. – Length, Size, width

Quantitative vs. Qualitative data

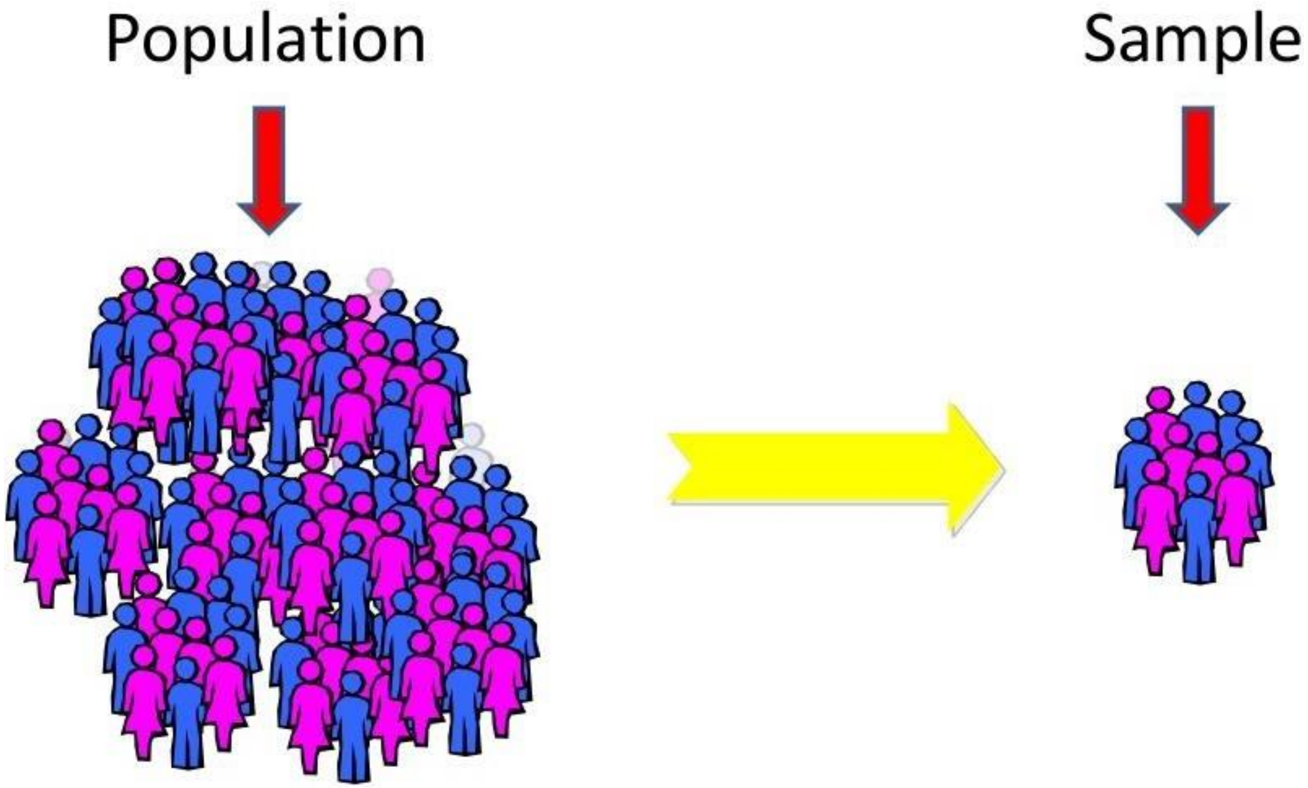
Data unit	Numerical variable = Quantitative data		Categorical variable = Qualitative data	
A Network Server	"How many **failed login attempts** were recorded today?"	1,240	"Which **operating system** is the server running?"	Linux (Ubuntu 22.04)
	"How many **open ports** were detected during the scan?"	5	"What is the **criticality level** of this asset?"	High (Production)
	"What is the **latency (ms)** of the connection?"	15 ms	"Is the **data encrypted** at rest?"	Yes (AES-256)

- Quantitative and qualitative data present different information, but both are used together to understand the complete picture of a problem

Time for thoughts and questions

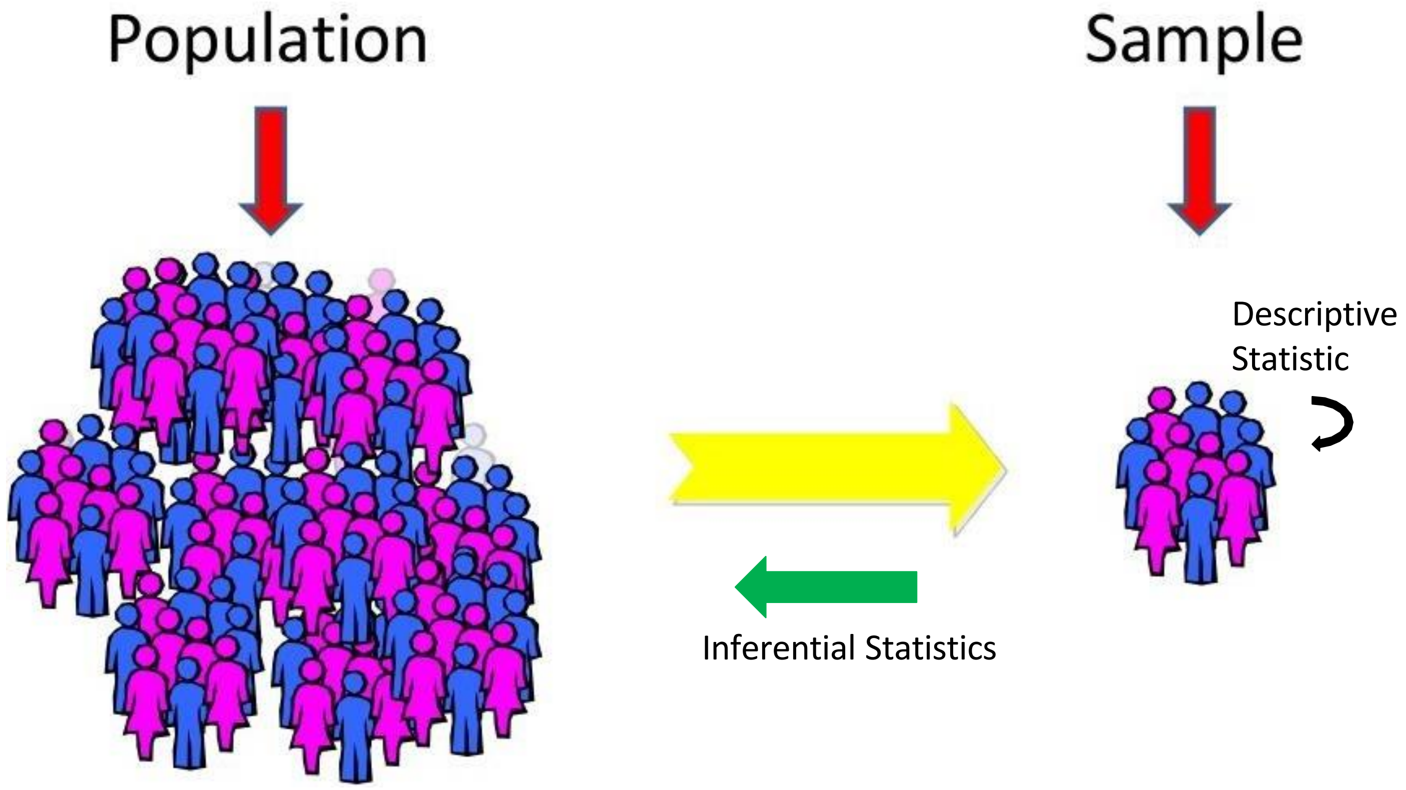
What is data distribution?

Population vs. Sample



Population vs. Sample

- **Descriptive Statistic** involves summarizing sample data in a meaningful way
- **Inferential Statistics** involves using the statistics from a sample to draw a conclusion about a population



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Data distribution

- A distribution is a function that displays all possible values for a variable and their frequency of occurrence.

- A frequency table displays data classes or intervals alongside the count of entries in each class.

Class width
$6 - 1 = 5$

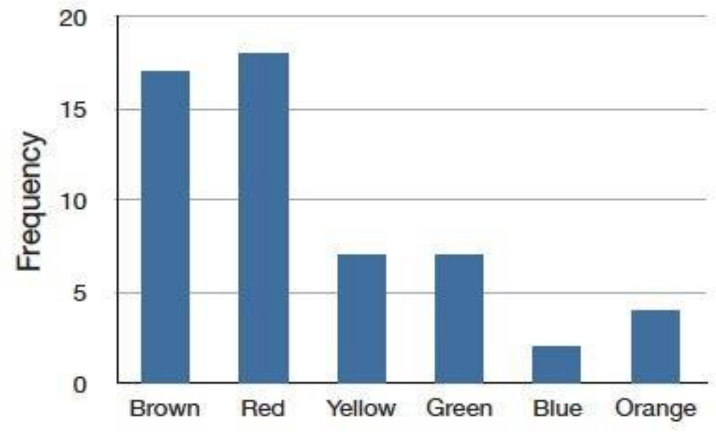
Class	Frequency, f
1 – 5	5
6 – 10	8
11 – 15	6
16 – 20	8
21 – 25	5
26 – 30	4

Lower class limits

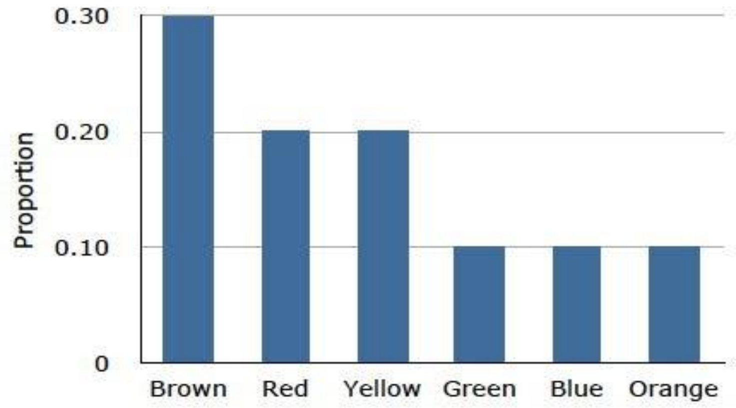
Upper class limits

Frequency vs. Probability Distribution

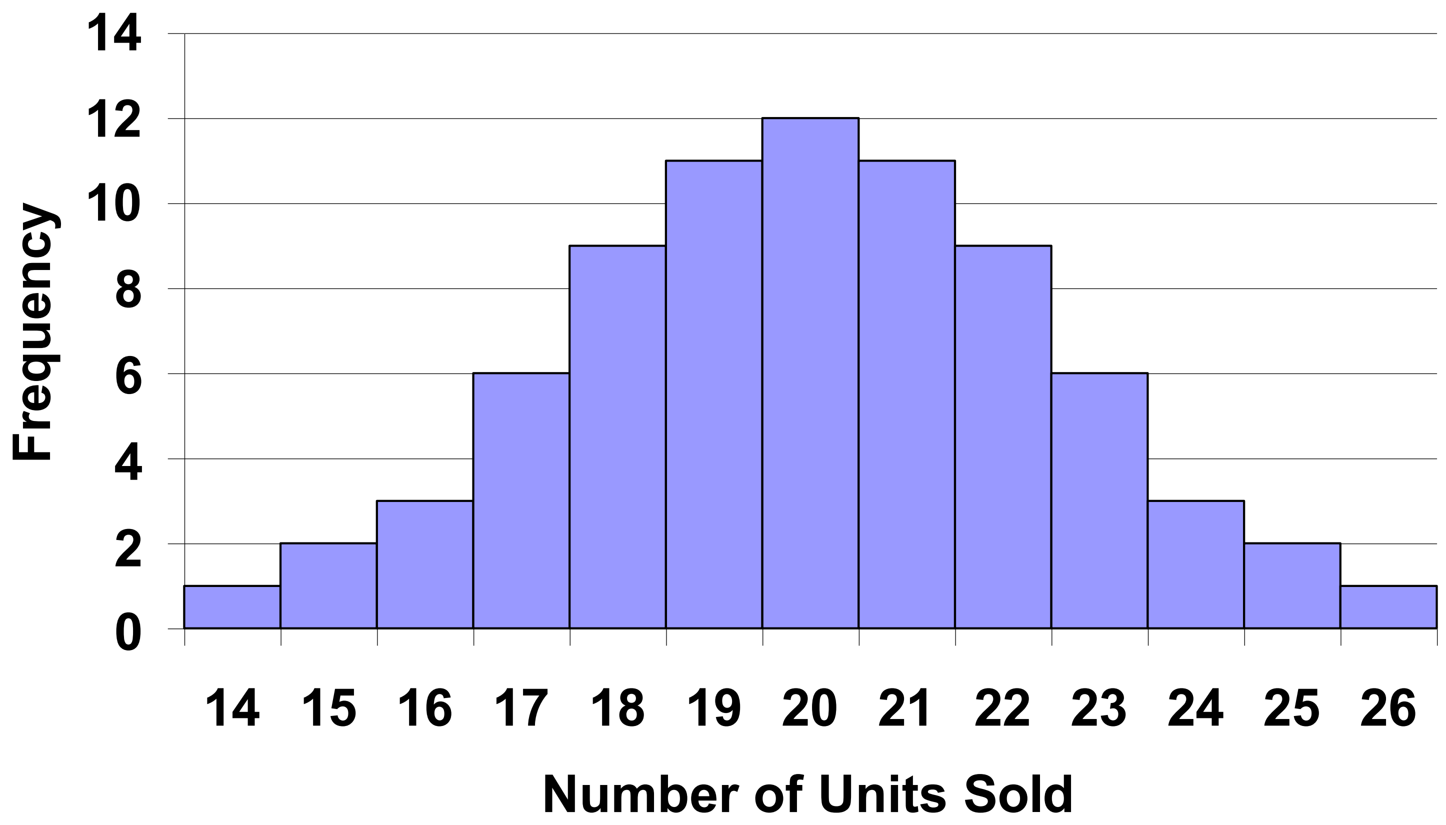
- A frequency distribution describes the occurrence of all values of discrete variables in the sample.



- A probability distribution defines the likelihood of each possible outcome for all variables in the data.



Normal (Symmetric) distribution



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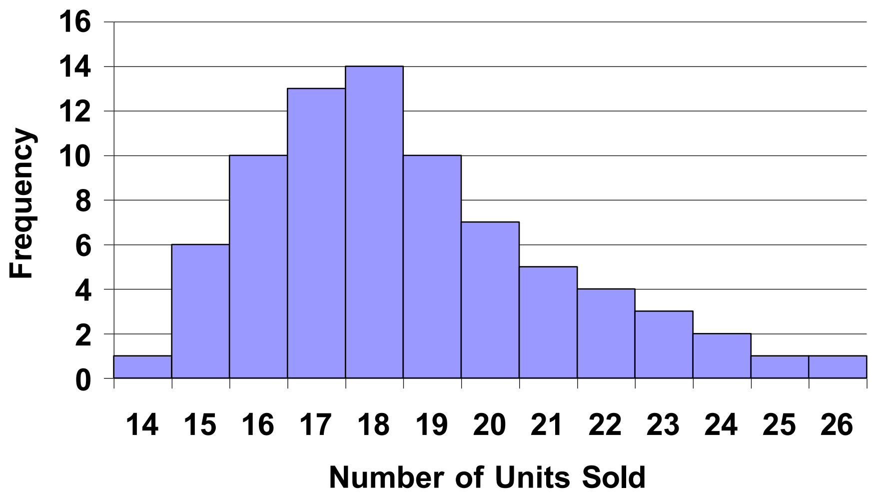
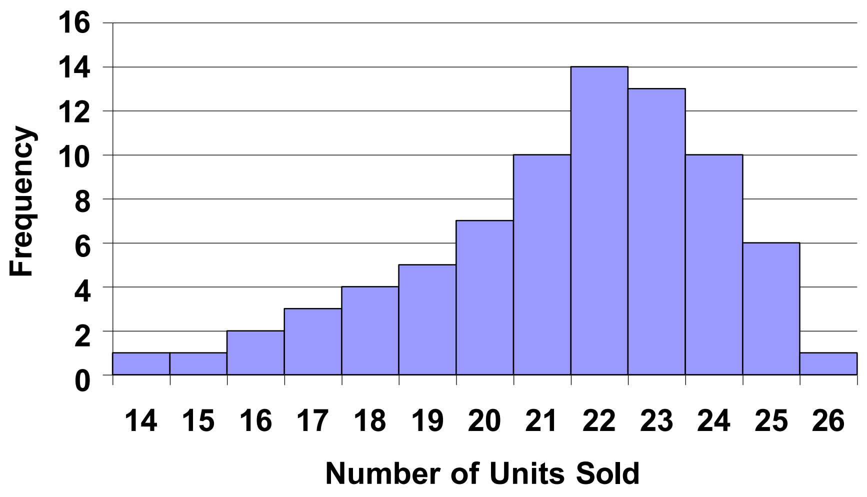
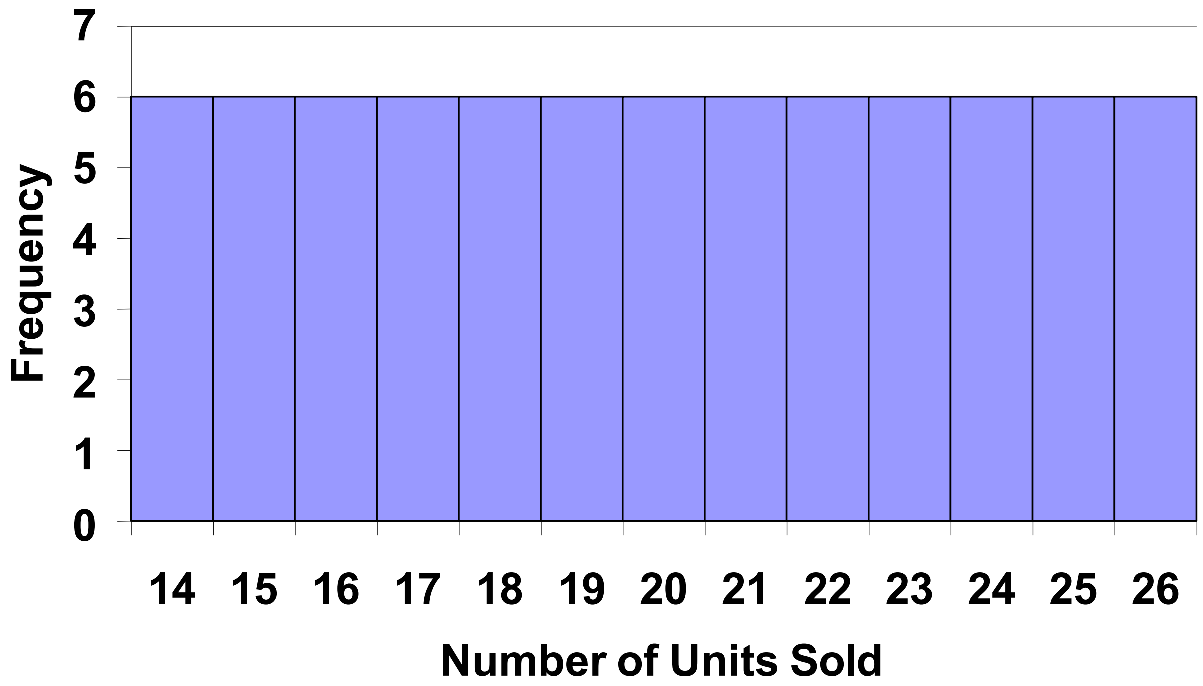
Skewed distribution





Uniform distribution



Time for thoughts and questions

Descriptive Statistics

Centrality and Variability Measures

How to describe data?

- Central tendency measures
  - describe the center around which the data is distributed
- Variation or variability measures
  - describe the data spread, i.e., how far the samples lie from the center

Centrality Measures: Mean

n = number of cases

x_i = ith case

f_i = ith case frequency

$\bar{x}$ = sample mean

- Sample mean of ungrouped data

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- Sample mean of grouped data

$$\bar{x} = \frac{1}{\sum_i^n f_i} \sum_i^n f_i x_i$$

Example - Incident Response Time

The SOC Manager at a financial firm wants to evaluate the response team's effectiveness in handling Unauthorized Access alerts. He gathered data on the Time to Containment (TTC), rounded to the nearest minute, from 50 recent security incidents.

45	38	53	27	45	32	59	40	57	22
31	29	32	49	26	35	39	35	32	36
64	34	22	28	57	65	37	25	40	33
69	45	57	48	28	43	28	31	29	27
34	22	42	58	61	39	65	39	29	48

- How can we describe this data?

Solution

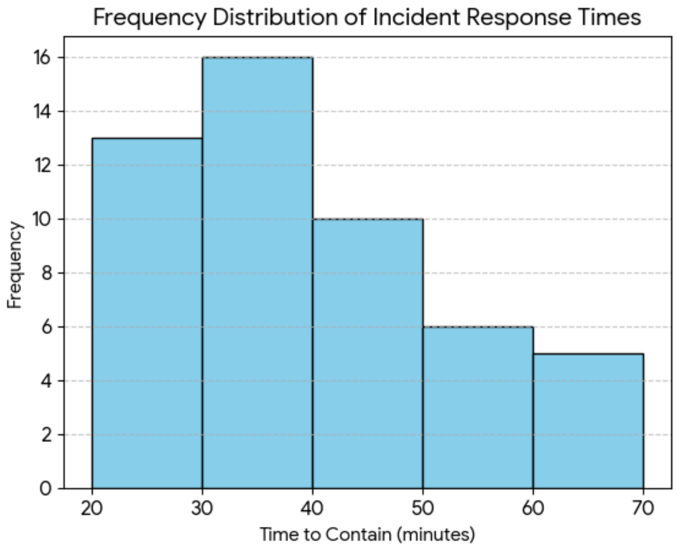
Raw Data

45	38	53	27	45	32	59	40	57	22
31	29	32	49	26	35	39	35	32	36
64	34	22	28	57	65	37	25	40	33
69	45	57	48	28	43	28	31	29	27
34	22	42	58	61	39	65	39	29	48

Tabular Summary

Time Range (min)	**Frequency**
20–29	13
30–39	16
40–49	10
50–59	6
60–69	5
Total	**50**

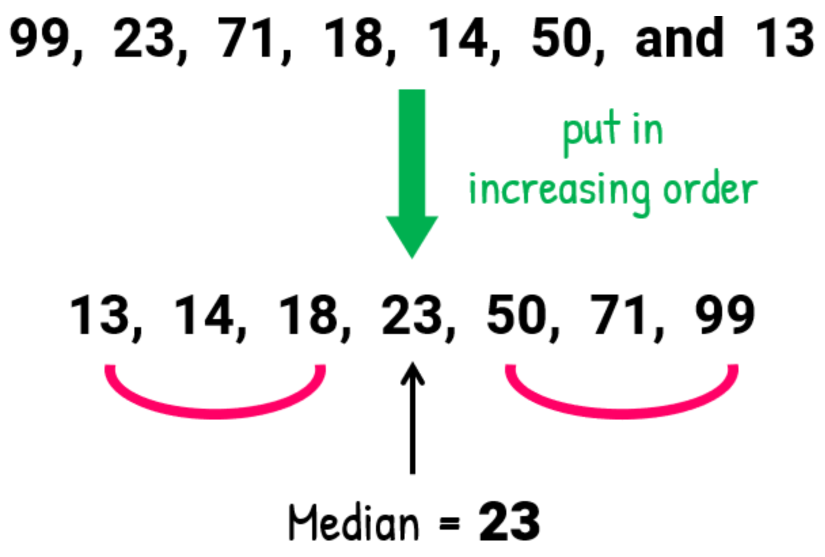
Graphical Summary



- Mean Time to Contain (MTTR) = total time taken to resolve an incident /number of incidents
  = 40.18 minutes
- This represents the average time taken to resolve an incident across all 50 samples.

Centrality Measures: Median

- Median is the exact middle value on a sorted list of data



- Median uses values in the center of the sorted data, thus not affected by extreme outliers

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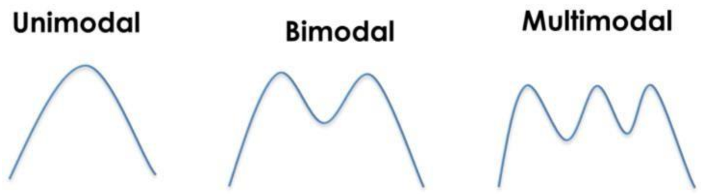
Centrality Measures: Mode

- Mode is used to describe most frequent value(s) in a sample
- Mode is not affected by extreme values, and it is suitable for qualitative data
- The mode may not be unique at times
  - For example,

- 2, 8, 5, 5, 4, 1, 9, 7, 3   Mode = 5
- 2, 2, 2, 4, 5, 6, 7, 7, 7   Mode = 2 or 7 (Bimodal)



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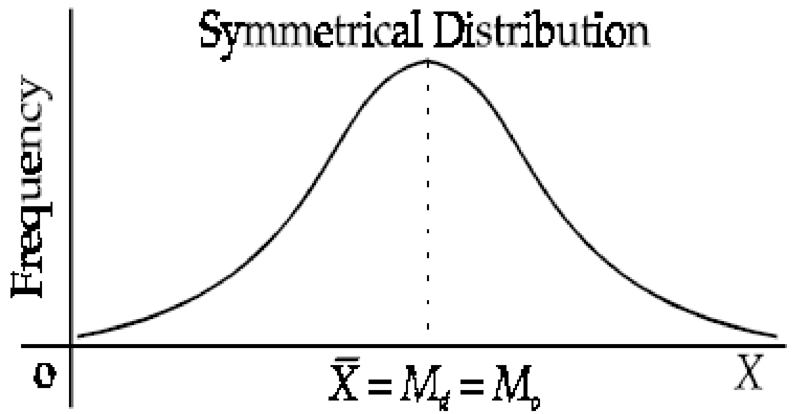
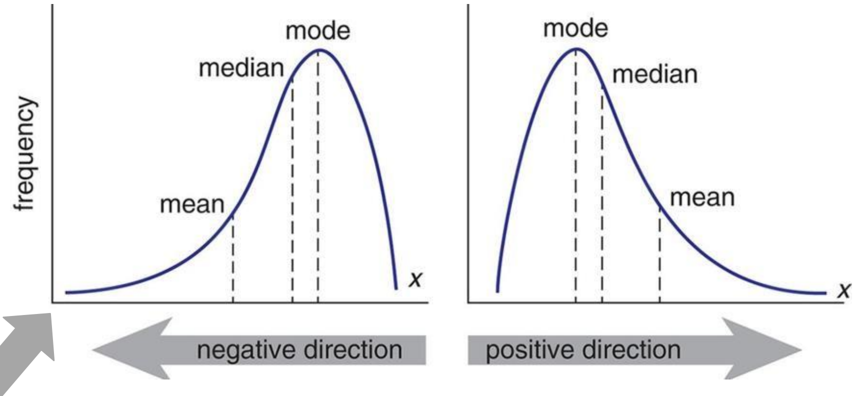
Relation between mean, median and mode

- Physically,
  - The mean can be interpreted as the center of gravity of the distribution
  - Median divides the area of the distribution into two equal parts
  - Mode is the highest point of the distribution
- For data that is symmetrically distributed, the mean, median, and mode can often be used interchangeably.





But not in these cases

Example 2

- Calculate the mean, median, and mode of the sample ages in the class. Determine which measure of central tendency best represents a typical entry in this data set. Are there any outliers present?

Ages in a class						
20	20	20	20	20	20	21
21	21	21	22	22	22	23
23	23	23	24	24	65	

Solution (1/2)

Ages in a class						
20	20	20	20	20	20	21
21	21	21	22	22	22	23
23	23	23	24	24	65	

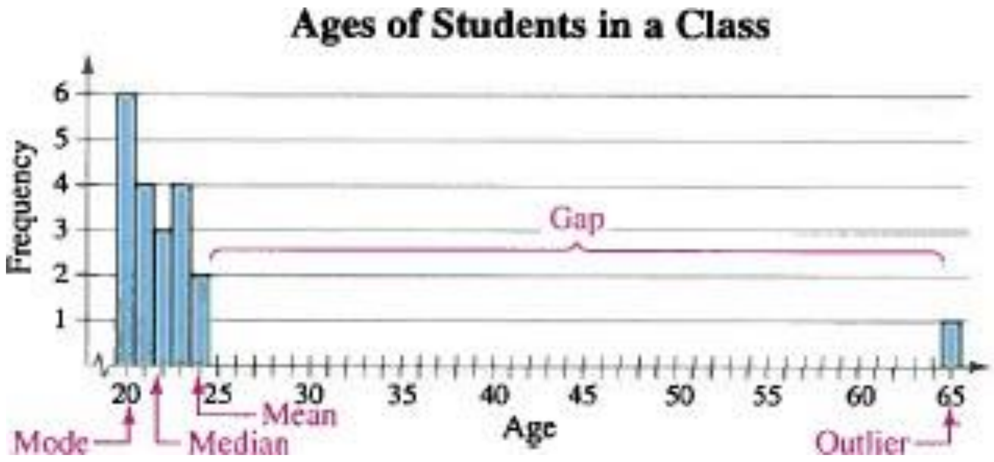
Mean: $\bar{x} = \frac{\Sigma x}{n} = \frac{20+20+\ldots+24+65}{20} \approx 23.8 \text{ years}$

Median: $\frac{21+22}{2} = 21.5 \text{ years}$

Mode: 20 years (the entry occurring with the greatest frequency)

Solution (2/2)

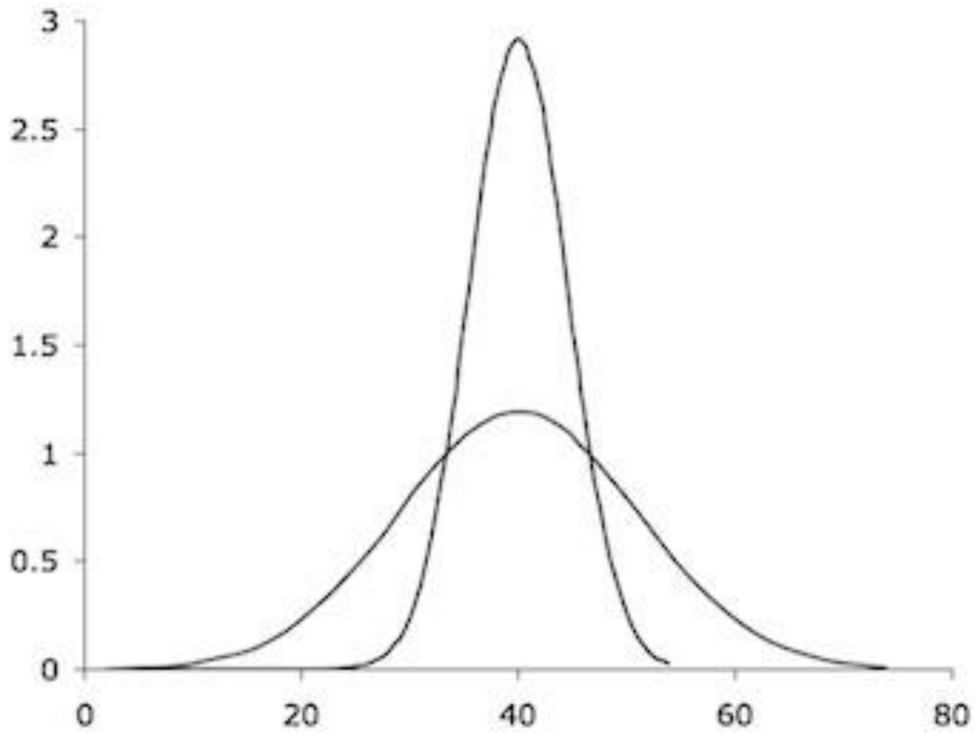
- Sometimes, a graphical comparison can assist in determining which measure of central tendency best represents a given data set.



- It appears that the **median** best describes the data

Dispersion and Variability

- Two data samples



- Different ways to measure the dispersion and variability of data
  - Mean Deviation
  - Variance
  - Standard deviation
  - Range

Estimate of Variability

Range: The difference between the maximum and minimum entries in the set

Example: A corporation hired 10 graduates. The starting salaries for each graduate are shown. Find the range of the starting salaries.

Starting salaries (1000s of NOK)

41 38 39 45 47 41 44 41 37 42

$$\text{Range} = (\text{Max. salary}) - (\text{Min. salary})$$
$$= 47 - 37 = 10$$

Estimate of Variability

n = sample size

x_i = ith sample

$\bar{x}$ = sample mean

s = sample SD

Mean Deviation (MD)

$$\text{MD} = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$$

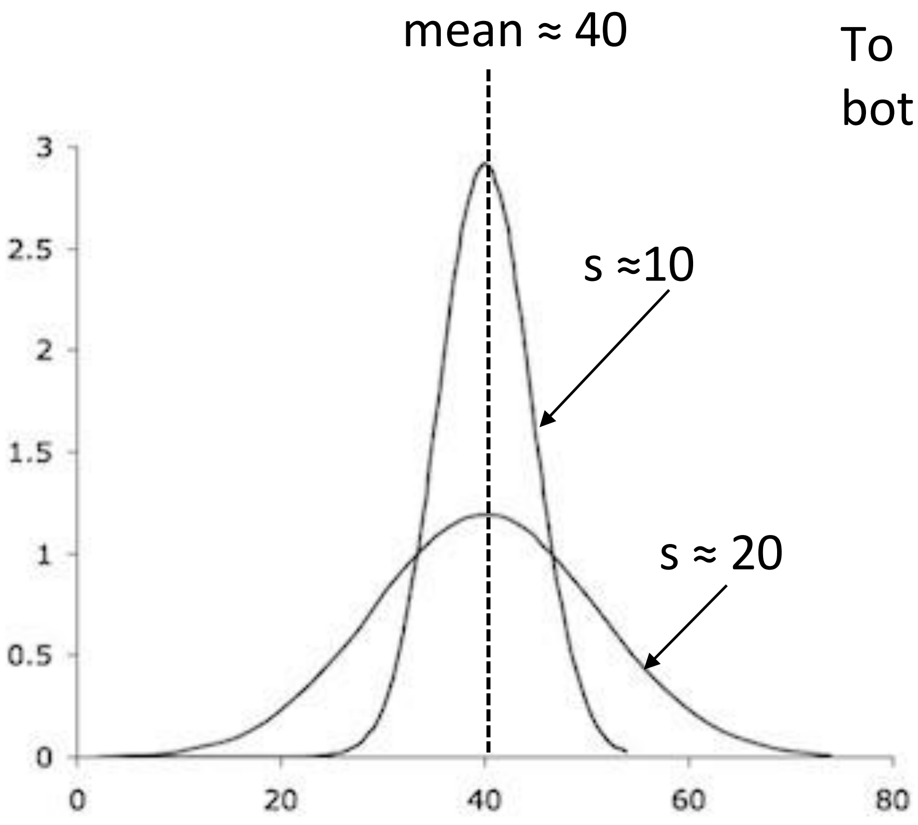
Variance (s^2)

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

Standard Deviation (s)

$$s = \sqrt{\text{Variance}}$$

Characterizing Distribution



To accurately describe a distribution, report both the mean and the standard deviation.

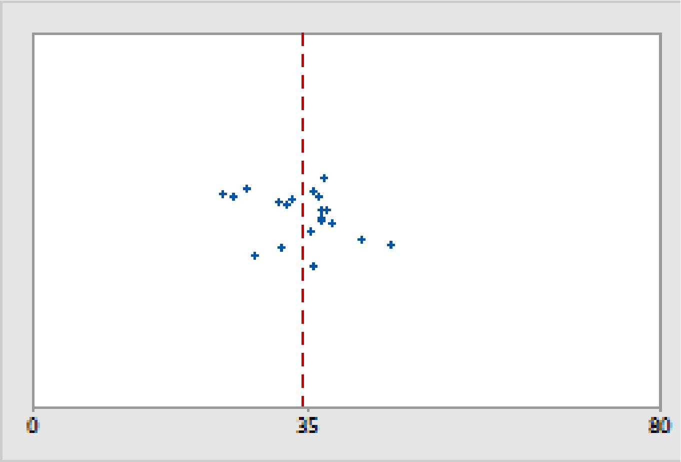
40 ± 10

40 ± 20

Example 3

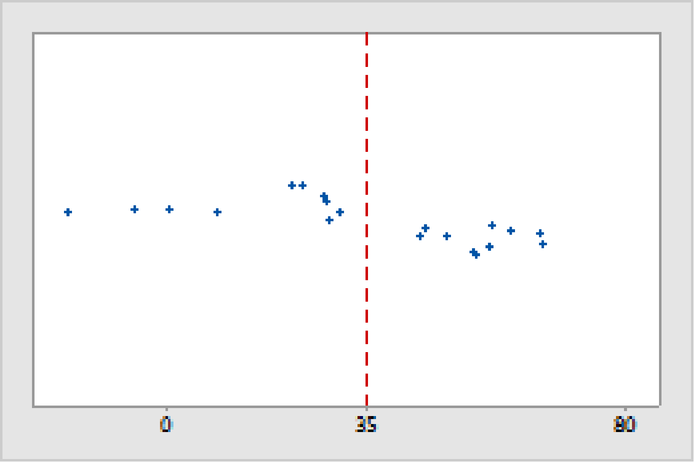
- Hospital administrators monitor the discharge times for patients treated in the emergency departments of two hospitals, A and B, to improve service quality. The plots below illustrate the distribution of the collected data.

Hospital A



Discharge time 35 ± 06 mins

Hospital B



Discharge time 35 ± 20 mins

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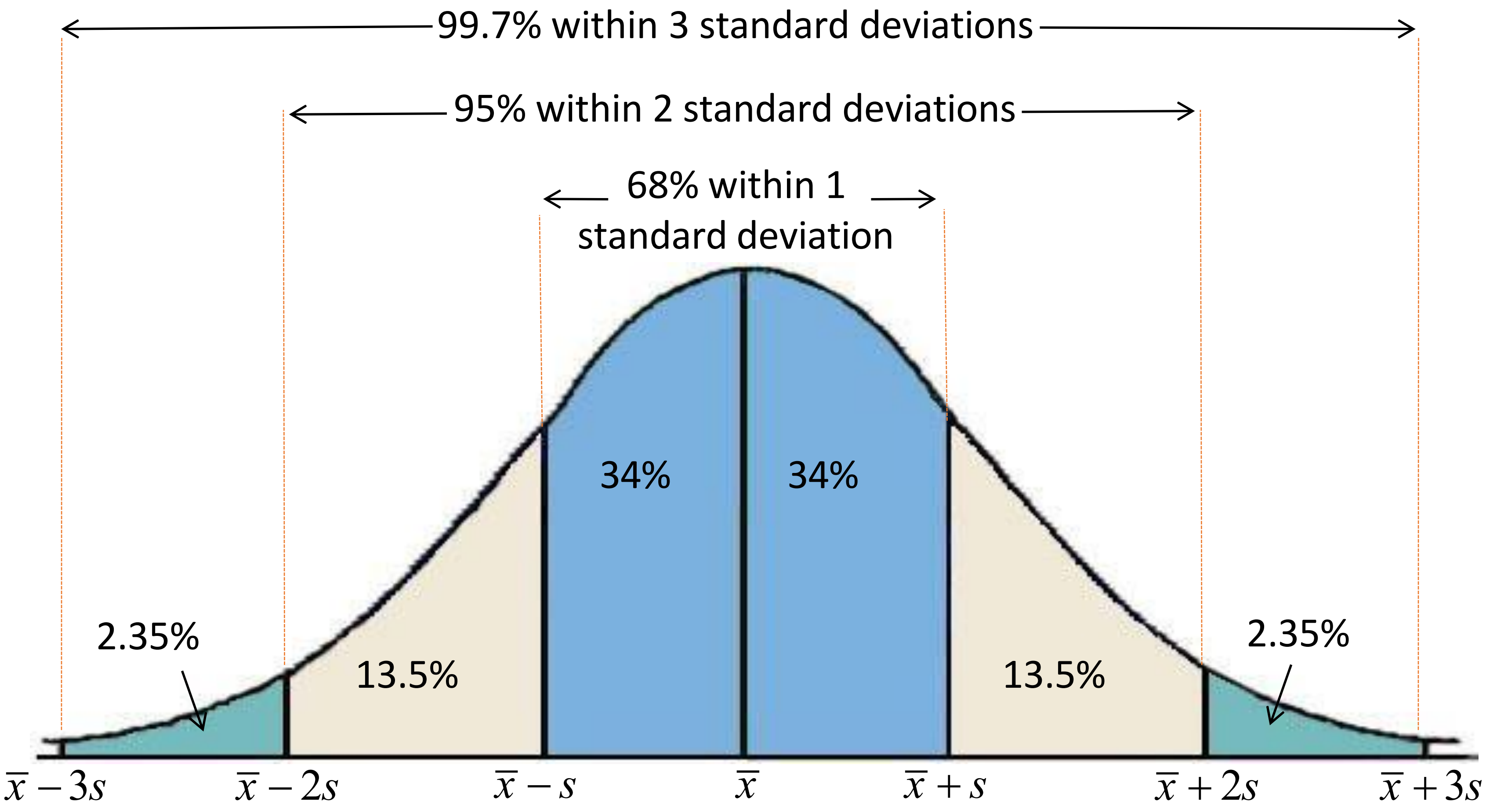
Empirical Rule (68 – 95 – 99.7 Rule)



Time for thoughts and questions

Visual Data

Visual data are used extensively in scientific research and in the media generally

Used to convey the information in the data

Two main reasons for using visual data

- Exploration
- Explanation

Exploratory vs. Explanatory

Exploratory analysis: to understand the data

- Discovering what might be noteworthy or interesting to highlight
- Exploring data is a much more individual matter

Explanatory analysis: to communicate the information

- Turn the data into information that can be consumed
- The visuals must be appropriate for the content and for the intended audience

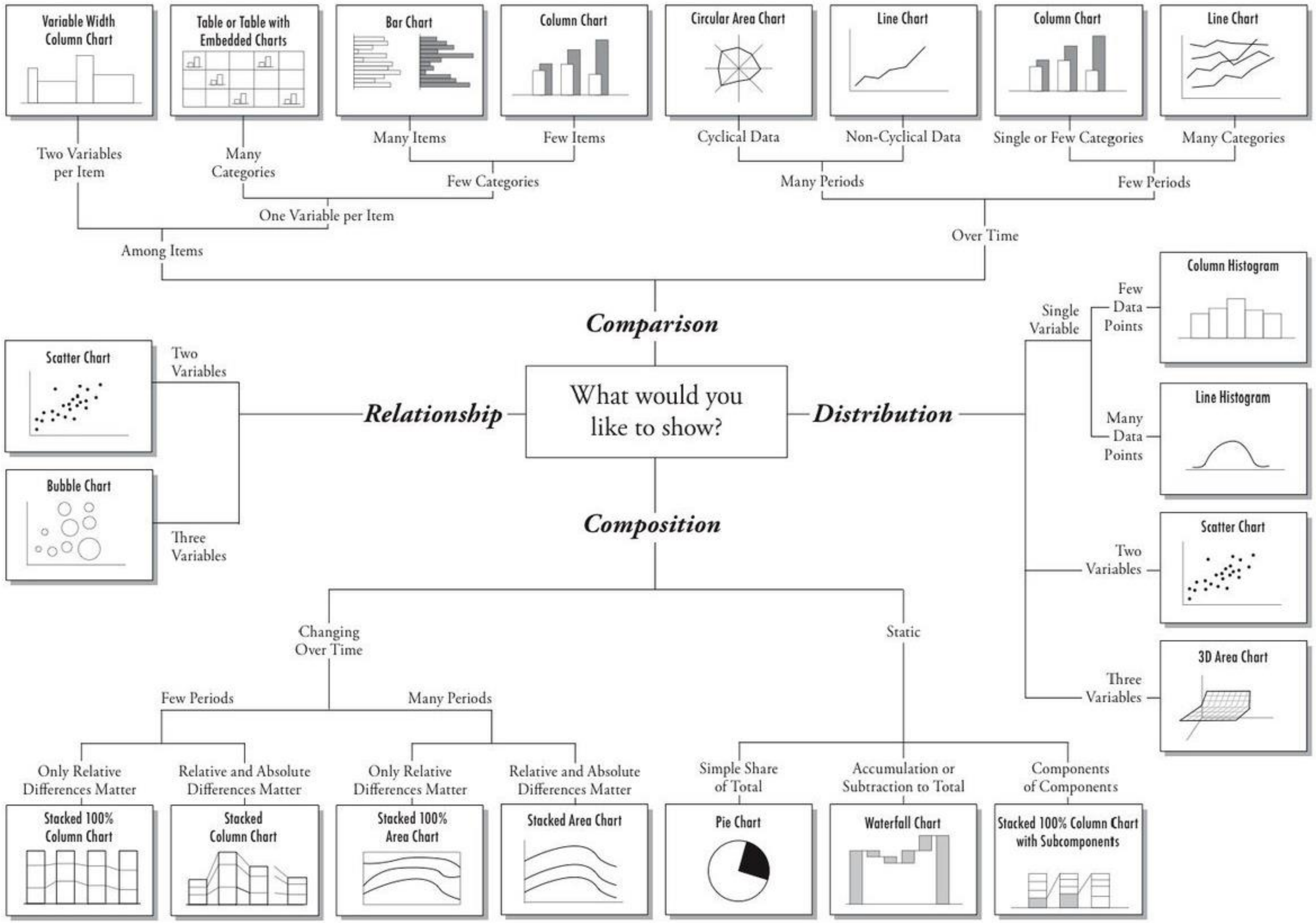
Choosing effective visuals

Choosing effective visuals

There are many types of visual out there

Select the type that will enable you to clearly get your message across properly

Use familiar types of visuals to enhance accessibility and understanding

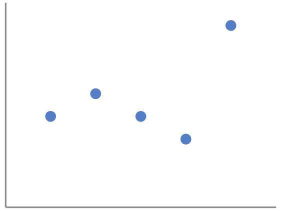


https://t.co/absN9ZUqW7

Commonly used visuals

91%

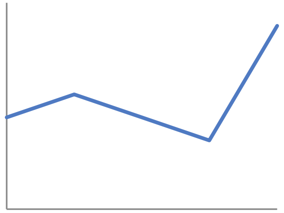
Simple text



Scatterplot

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

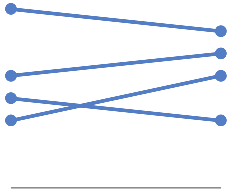
Table



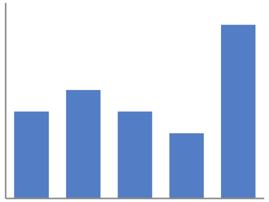
Line

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Heatmap

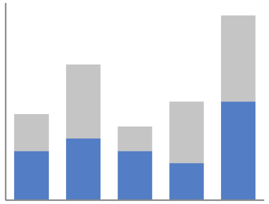


Slopegraph

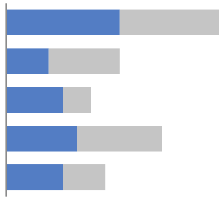


Vertical bar

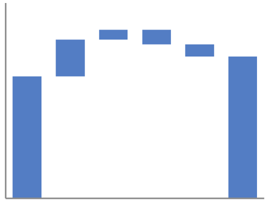
Horizontal bar



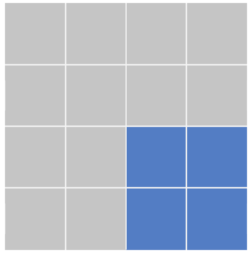
Stacked vertical bar



Stacked horizontal bar



Waterfall



Square area

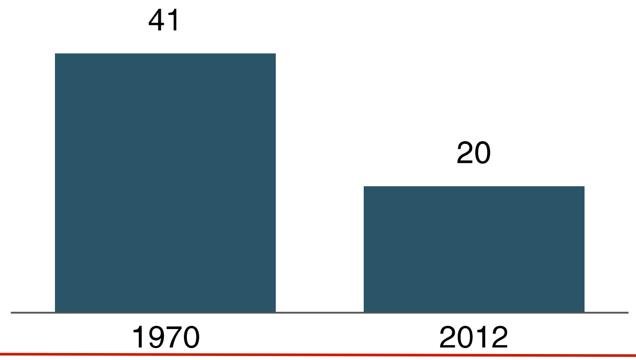
Simple text

Simple text can be a great way to communicate a number or two

Example

Children with a "Traditional" Stay-at-Home Mother

% of children with a married stay-at-home mother with a working husband



20%

of children had a **traditional stay-at-home mom** in 2012, compared to 41% in 1970

Table

They can be used to communicate multiple different units of measure

Table should be design to fade into the background, letting the data take center stage

Heavy borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Light borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Minimal borders

Group	Metric A	Metric B	Metric C
Group 1	$X.X	Y%	Z,ZZZ
Group 2	$X.X	Y%	Z,ZZZ
Group 3	$X.X	Y%	Z,ZZZ
Group 4	$X.X	Y%	Z,ZZZ
Group 5	$X.X	Y%	Z,ZZZ

Heatmap

A way to visualize data in tabular format

Use colored cells to convey the relative magnitude of the numbers

Heatmap

LOW-**HIGH**

	A	B	C
Category 1	15%	22%	42%
Category 2	40%	36%	20%
Category 3	35%	17%	34%
Category 4	30%	29%	26%
Category 5	55%	30%	58%
Category 6	11%	25%	49%

Graphs

While tables interact with our verbal system, graphs interact with our visual system

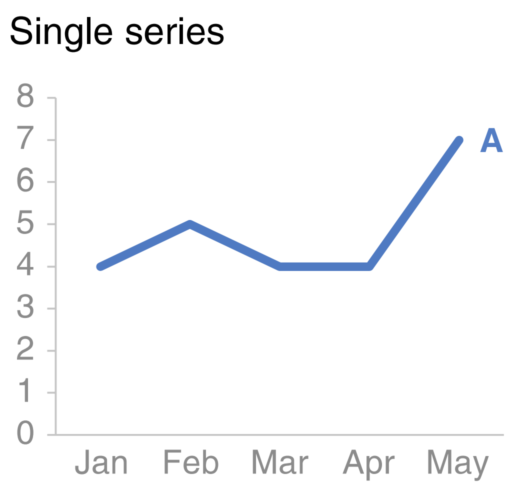
A well-designed graph will typically get the information across more quickly than a well-designed table

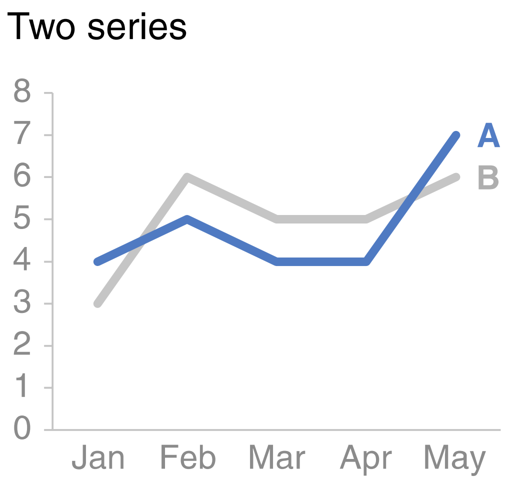
Distinction between charts and graphs?

Line graphs

Line graphs are mostly used to plot continuous data

- Points are physically connected via the line
- X-axis represent the time
- Y-axis the magnitude of the value





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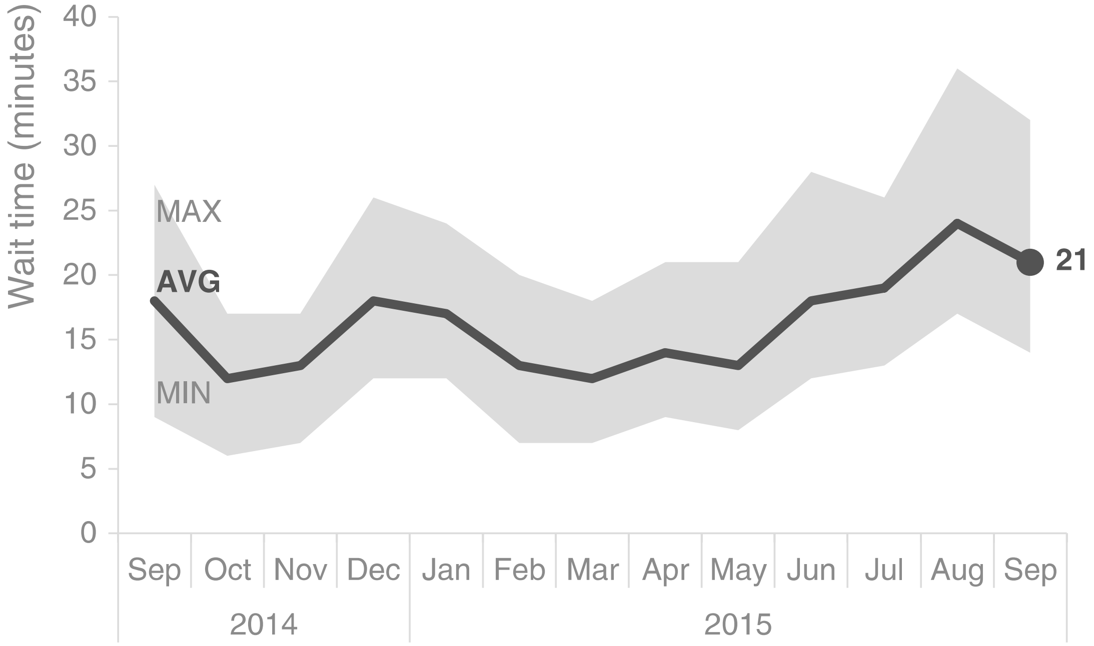
Line graphs

Line may represent a summary statistic, like the average, or the point estimate of a forecast

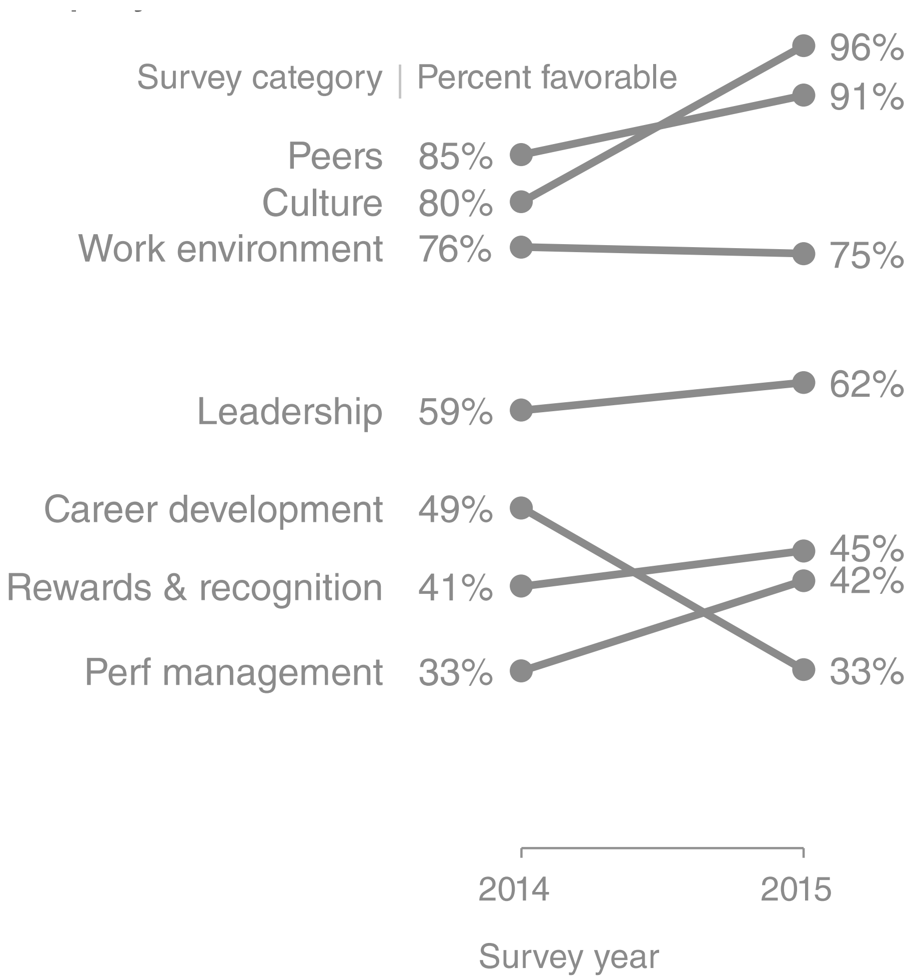
It is good to give a sense of the range, directly on the graph by also visualizing this range



Passport control wait time - Showing average within a range

Slopegraph

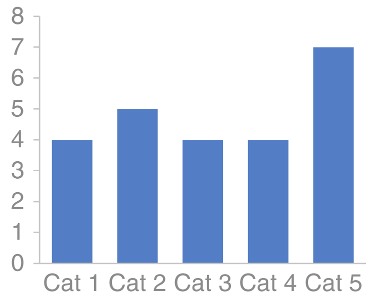
Used to show relative increases and decreases across various categories between the two data points



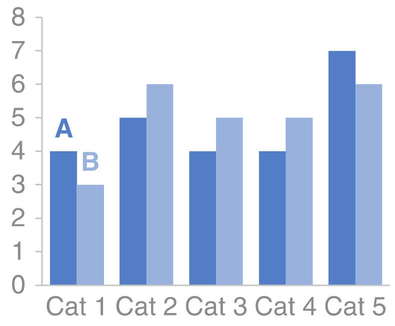
Bar chart

Bar charts are easy for our eyes to read and are used to compare data

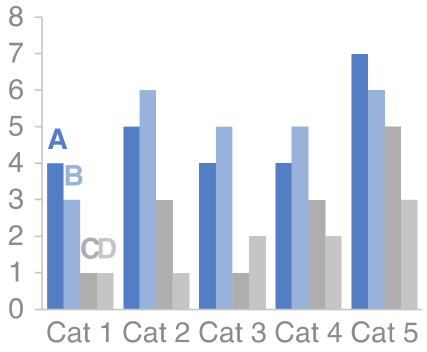
Single series



Two series



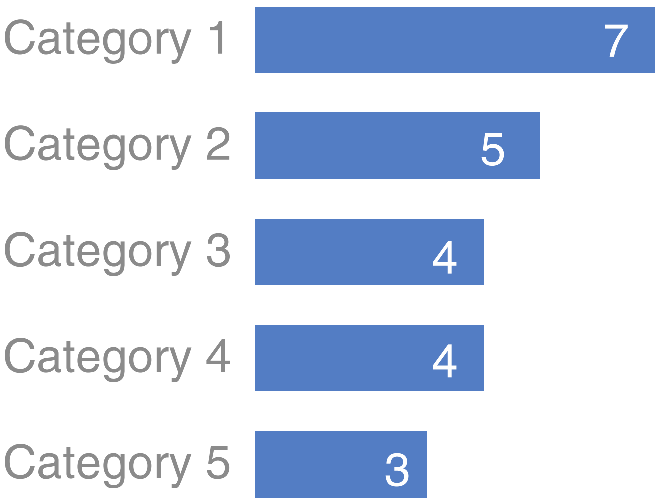
Multiple series



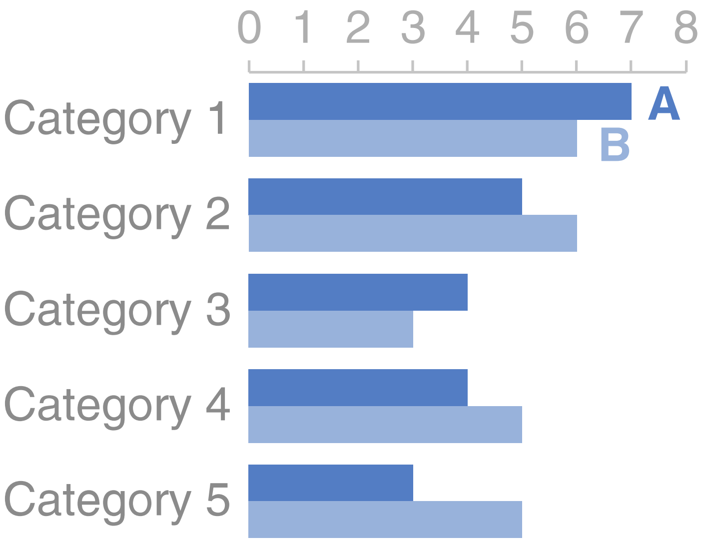
Bar chart

Horizontal bar chart

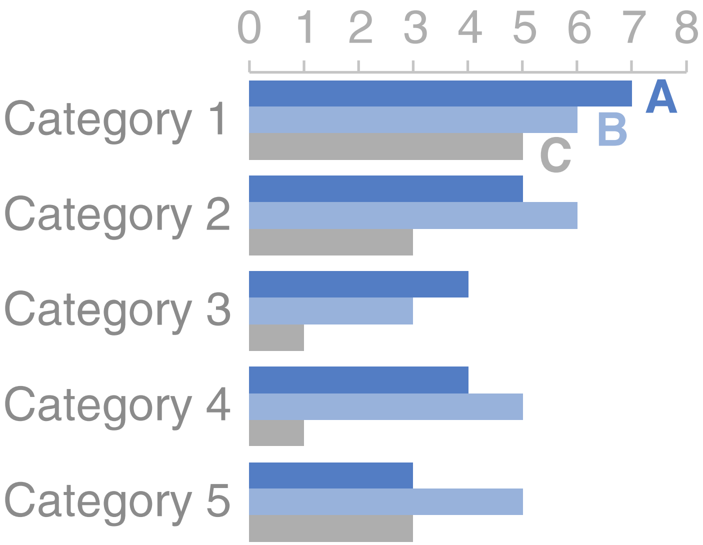
Single series



Two series

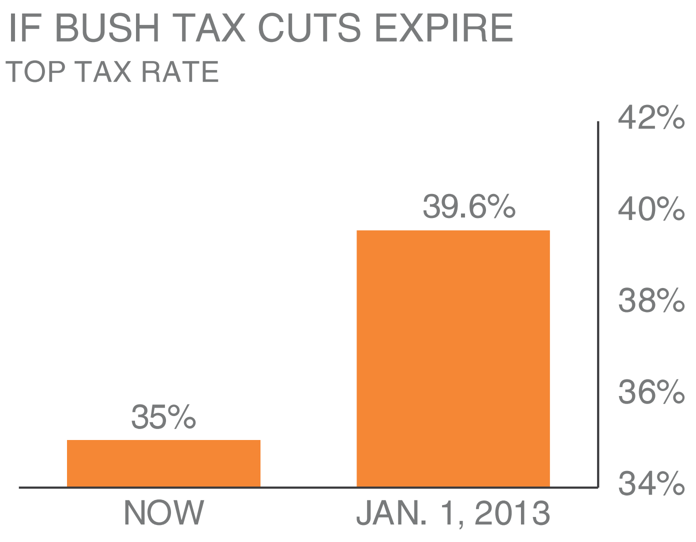


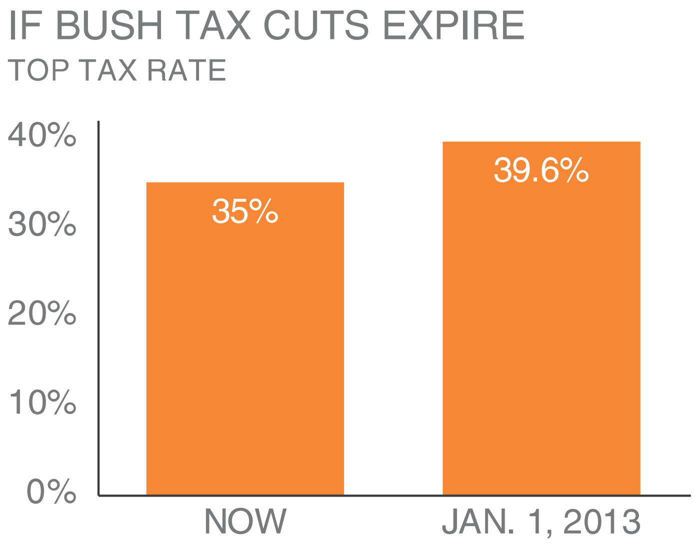
Multiple series



Bar Chart

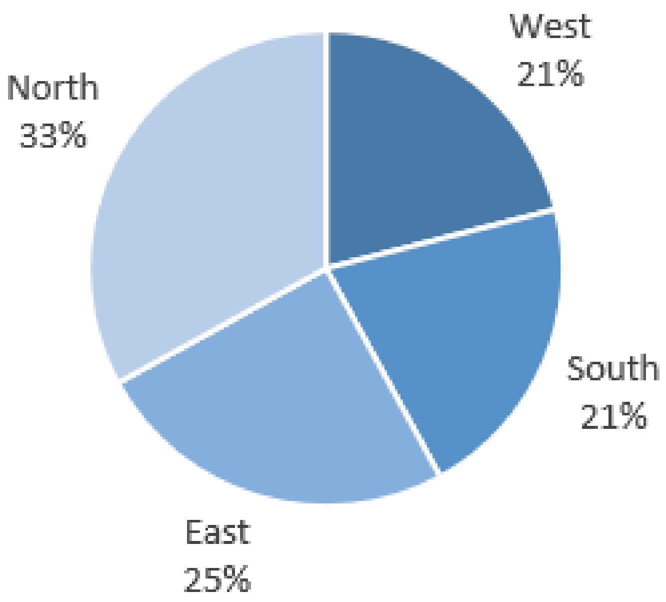
To avoid causing visual ambiguity it is important that bar charts always have a zero baseline (i.e., **x**-axis crosses the **y**-axis at zero)





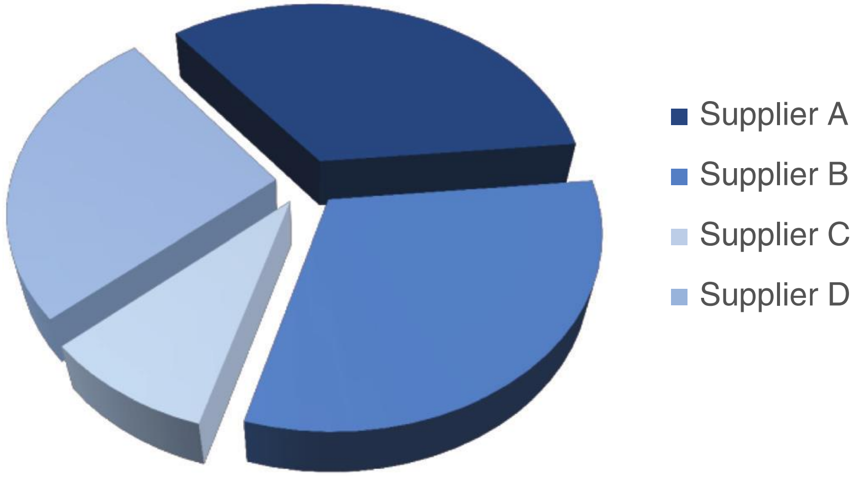
Pie Chart

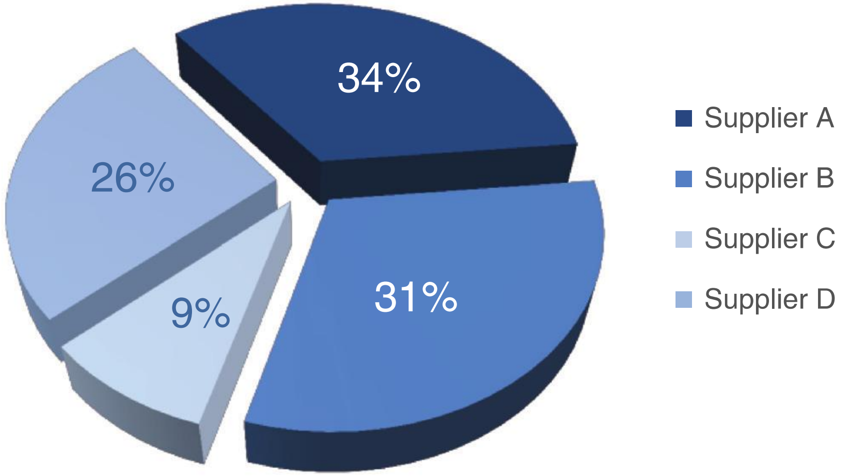
Shows how categories represent part of a whole-composition

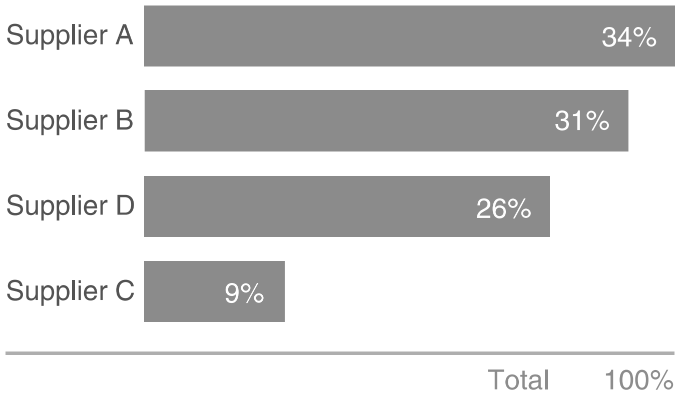


Pie Chart

3D pie chart



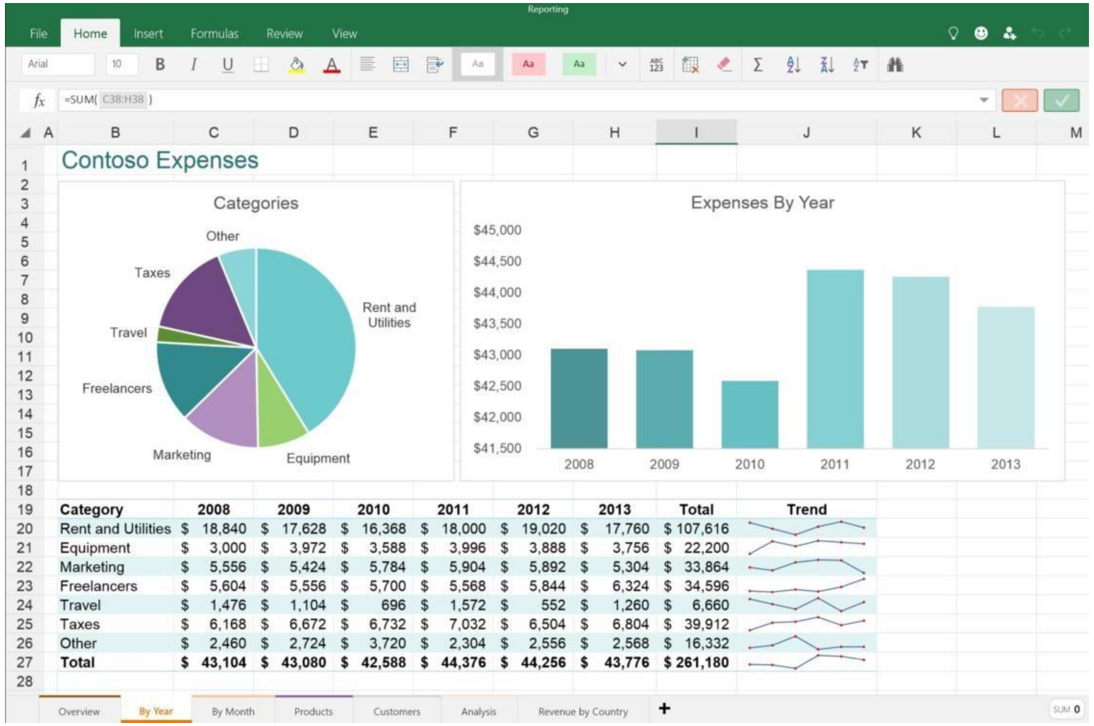




Don’t use 3D! It does nothing good; it skews the visual perception of the numbers

Tools for statistical analysis

- Programing software
  - Python: libraries and features for data analysis
  - R: programming language of statisticians
- Non-Programing tools
  - Excel: bread and butter tool for data exploration
  - SAS: statistical analysis software
  - SPSS: statistical package for the social science



Tools for data analysis

- Computational Statistics in Python

https://people.duke.edu/~ccc14/sta-663-2017/

- Basic Statistics using Microsoft Excel

https://mathcs.org/statistics/course/index.html

Time for thoughts and questions

Noroff University College